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Visual Interface / Design

Logo

As a logo wasn't provided in the set task brief, I took it upon myself to design / create a logo for Rolsa Technologies.

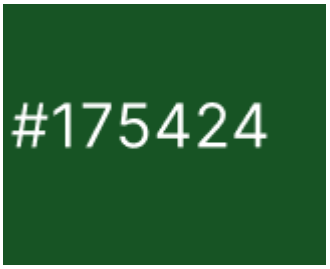
The logo I have come up with is a simple, yet effective design with 3 variations depending on the situation / need.

This is the first logo. It uses a green color scheme to signify "green" energy, using common appliances within the logo that resemble items that make the user think of green energy such as electricity, or solar panels.
The second and third variations are the same logo, however, they are split / cropped to have their own sections. This is so that they can be used in different ways, such as just the image for the header or just the text in places where an image is unnecessary.

Color Scheme

Using a website called coolors (www.coolors.co), I generated a random color pallet until I found one that would fit my design. In the end, I settled upon this color pallet:

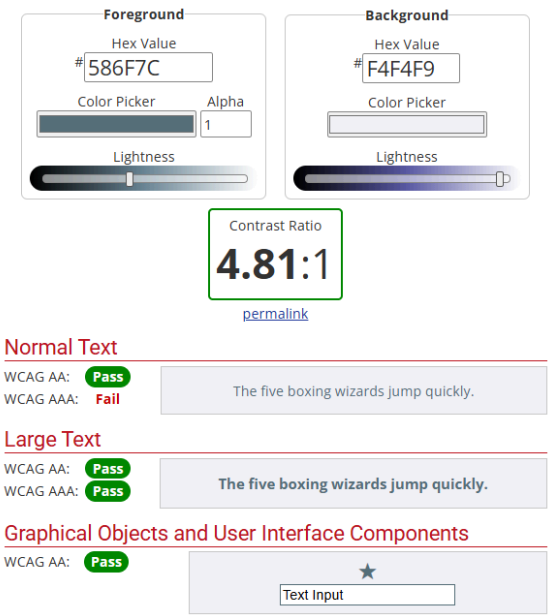
(#000000, #2F4550, #586F7C, #B8DBD9, #F4F4F9)
This is because it has high contrasting colors, and I also figured that I could implement my own color if needed, and that is what I did.
When creating my wireframe, I found little use of the color #B8DBD9 (light blue), and so, I opted to use the green that I used within my logo.

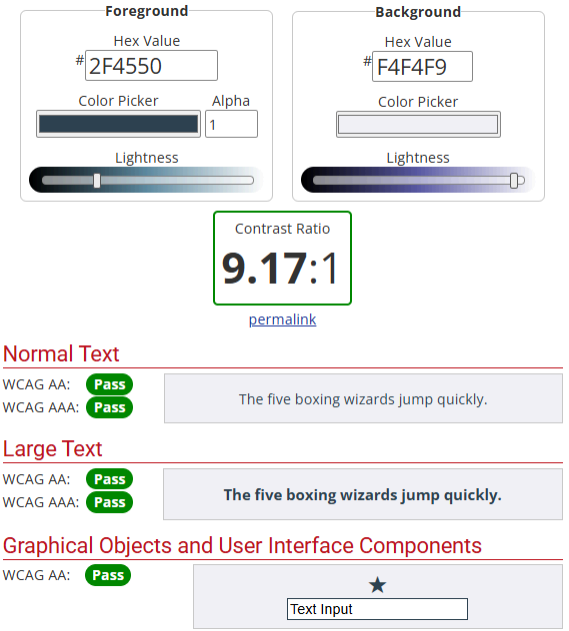


This turned out to be the perfect color as it has a high contrast value with the text (as you'll see later) and it represents green energy or technologies well.

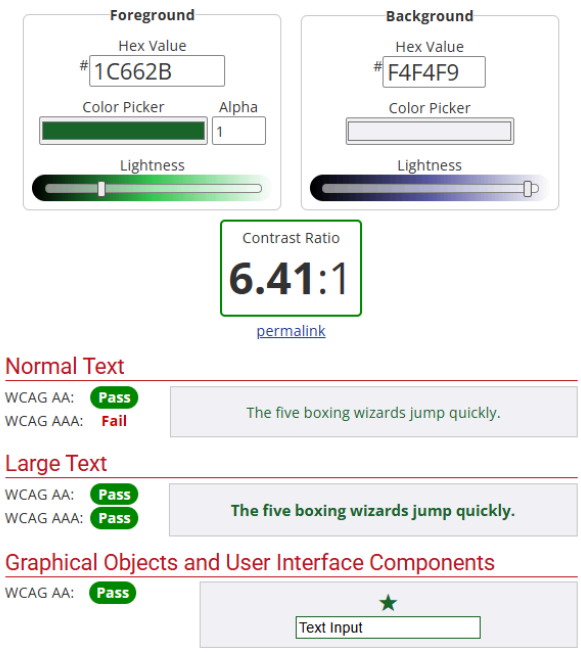
Using the website WebAIM (Web accessibility in mind), they provide a contrast checker that is free to use (https://webaim.org/resources/contrastchecker/). I used this to ensure that my colors have good contrast values so that they would be easy to read. I originally thought that i would use the value #586F7C as my text color, however, after comparing it to the value of my background color (#F4F4F9), the contrast failed at one test.

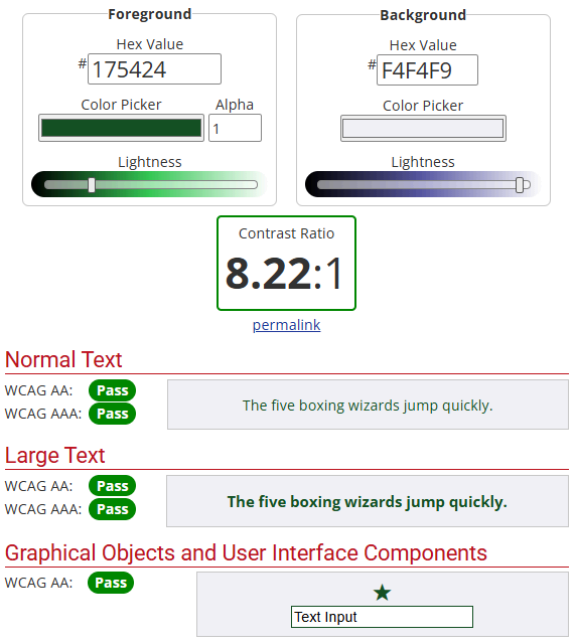
In order to fix this, I changed my text color from my original #586F7C, to the slightly darker #2F4550. This made the contrast ratio jump up and make it much better, passing all the tests.





Following this, my original green color has a value of #1C662B, but after doing the tests on my background color, it failed at the very same test. Due to this, I decided to make the green slightly darker





These two colors are the colors I’ve since used across my entire page for my text.

Design

In the following pages, I'll be showing off the wireframe I have designed for this solution. Some aspects of the wireframe are bound to change in the development process as new ideas are implemented / some may have been forgotten during the design process. Additionally, the images and text are bound to change completely as they are currently just placeholders. These images are unlicensed therefore are not allowed to be used in a commercial / final product.

Main Page

Navigation Bar and Hero Image

The main website consists of one page, split into different sections for the different areas that will be implemented. There will be external pages for more information or the account section of the website.

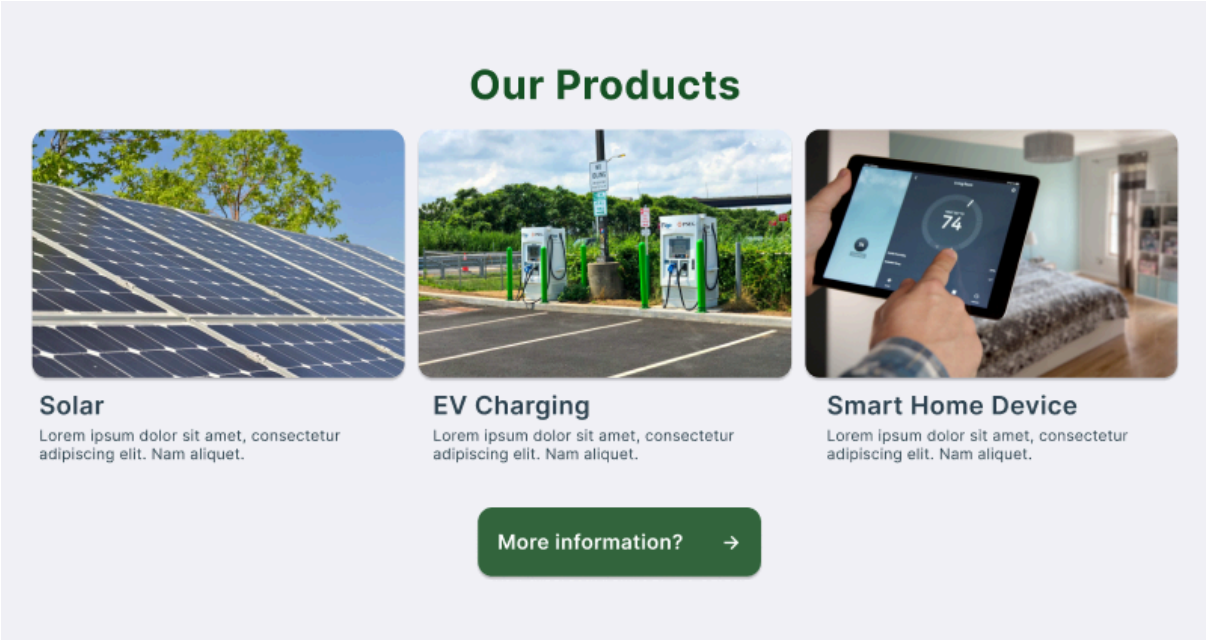
The first section will consist of a hero image with the company logo and a short catch phrase. This is so that the user gets instantly hooked once they enter the website. I also decided to add a little transparent background (the white and black boxes) behind the text and logo so that they are more easily visible for the user. The navigation bar will be consistently on the top as the user scrolls down, allowing for easy navigation.

About us section

The second part of the solution is the about us section. It is currently filled with placeholder lorem ipsum text as there was not much detail provided about what the company does. This will likely remain the case as further communication with the company would be necessary to figure out what exactly the company does, allowing for accurate text. Additionally, it is also really easy to configure what the actual text says. In short, this section will be a short description about the company's history and what they do to engage the user.

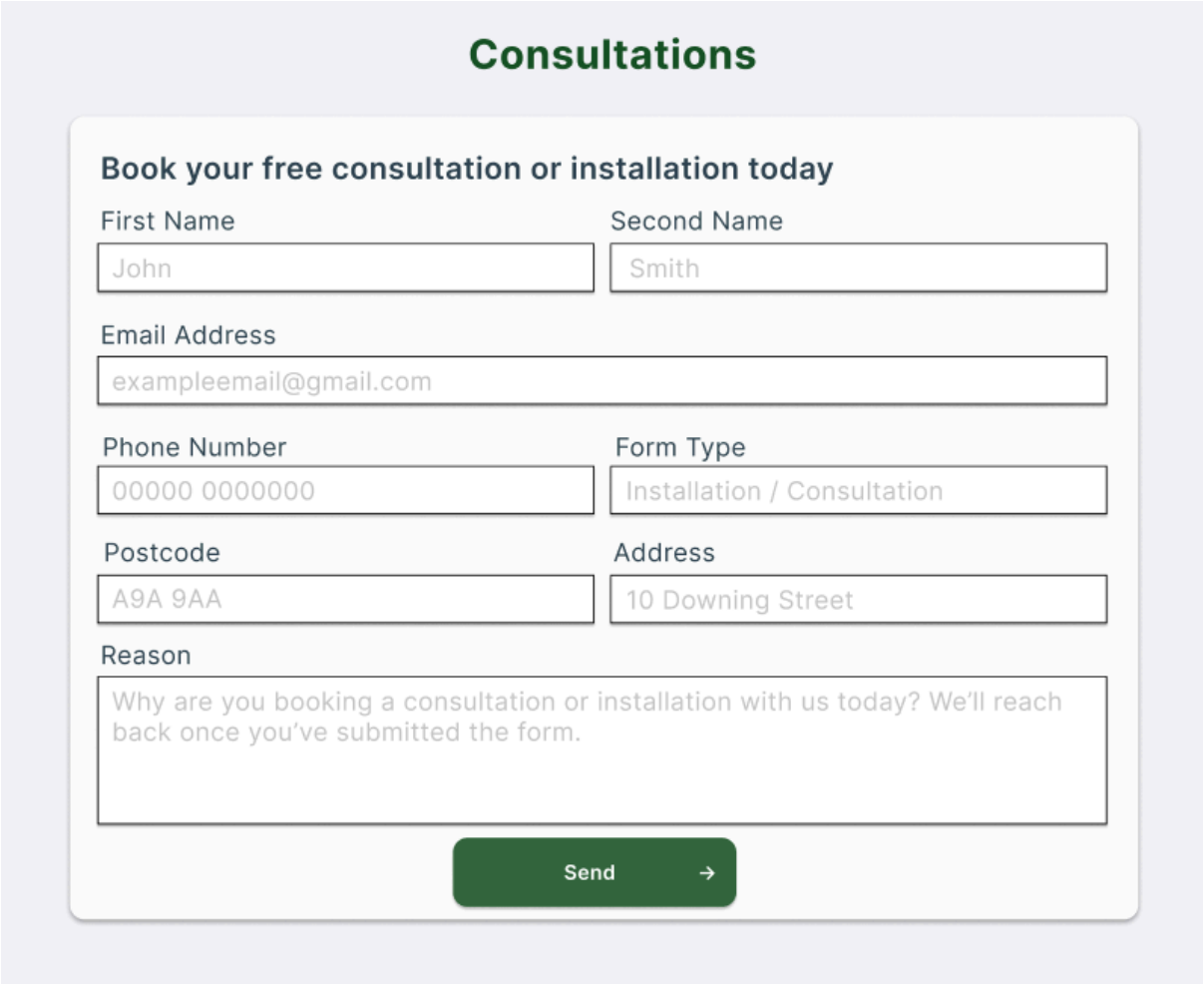
Our Products section

The 3rd section is the “our products” page. This section will contain short descriptions of what the company provides (the 3 main services mentioned in the brief), as well as the possibility of exploring more of what the company does by pressing the more information button. This will allow users to explore further information regarding the specific product / service, allowing the user to make a fully justified decision about whether or not they wish to purchase a product..



Consultations / Installations section

The fourth section is a simple form for consultations or installations.
As the company provides both, for simplicity and ease of use, I've merged the different sections into one, allowing the user to book a consultation or installation online in one place. Confirmation will then be sent either via telephone or email to the user within the later stage, and payment will be processed if necessary (This is for future development purposes, not to be integrated into this version as no valid platforms are available to set this up, such as local servers or email servers).

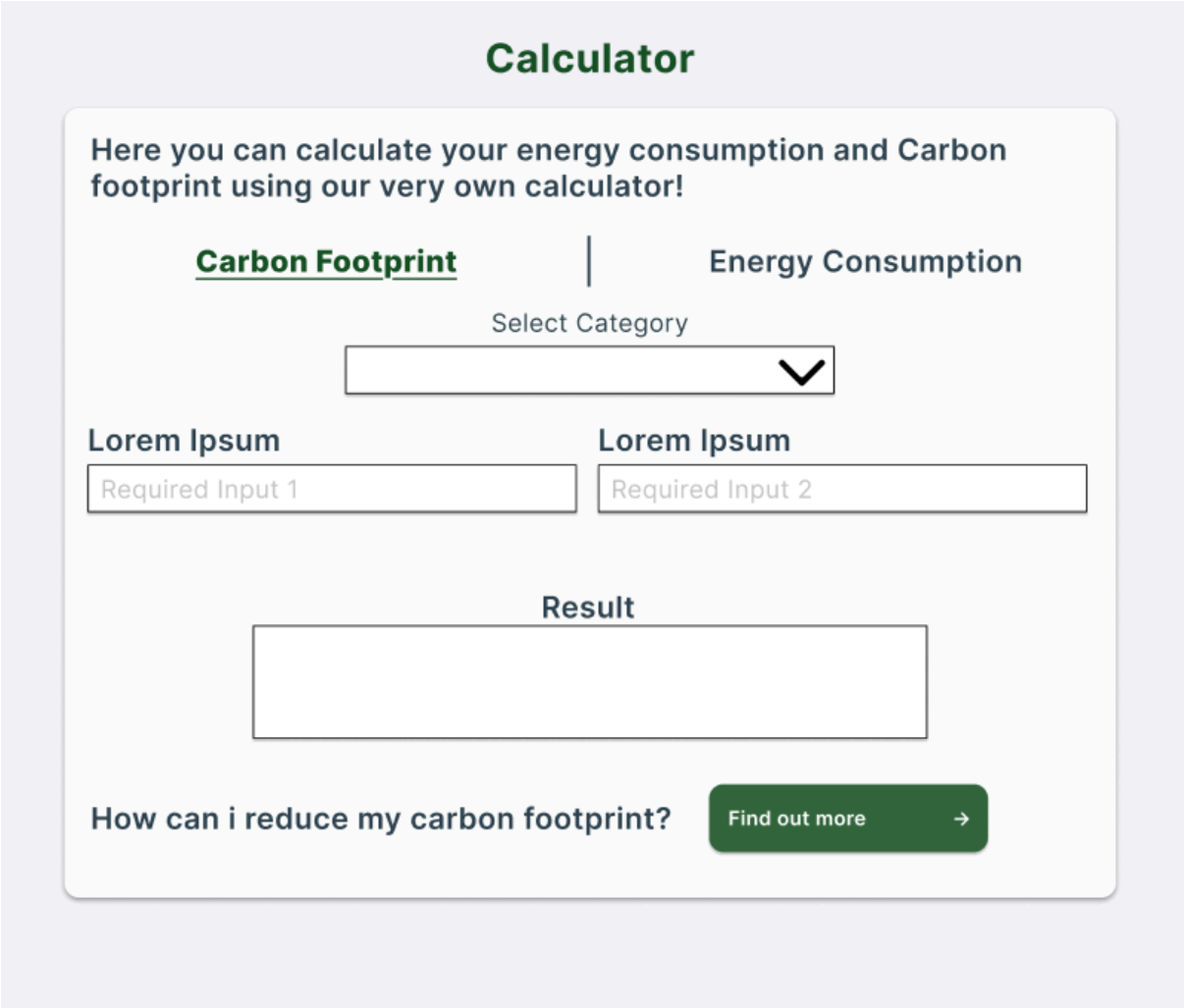


Calculator Section

The 6th section is an all in one calculator which allows the user to track and calculate either their carbon footprint or energy consumption.

This works by first either selecting the Carbon Footprint or Energy consumption option, which is highlighted in green and underlined, then proceeding to allow the user to select a category that depends on the prior option. For example, if carbon footprint is selected, the user will then be allowed to calculate from categories such as transport or food, whereas if energy consumption is selected, they'll be allowed to calculate their energy consumption within the month (refer to maths shown in the proposal). There are currently 2 placeholder input boxes for the calculation, but this can be edited for however many inputs needed for a result to be calculated.

Additionally, if the user presses the "Find out more" button, they will be prompted with an on screen UI that tells them how carbon footprint is measured, and what they could do to reduce their carbon footprint.



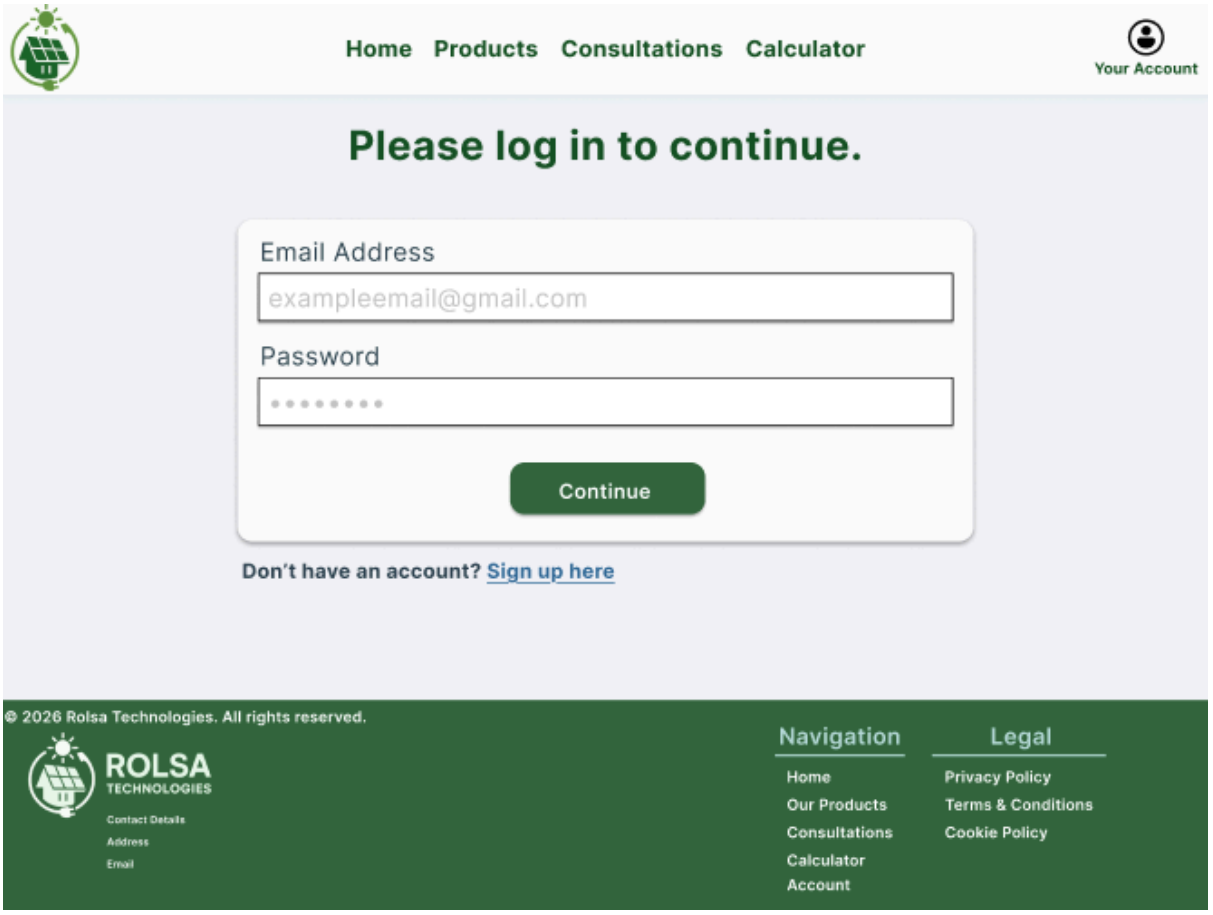
Footer

Finally, the footer.
It contains important information such as any legal policies, and copyright.
It also has spots for the address, email and contact details that can be filled in by the company.

Account Page

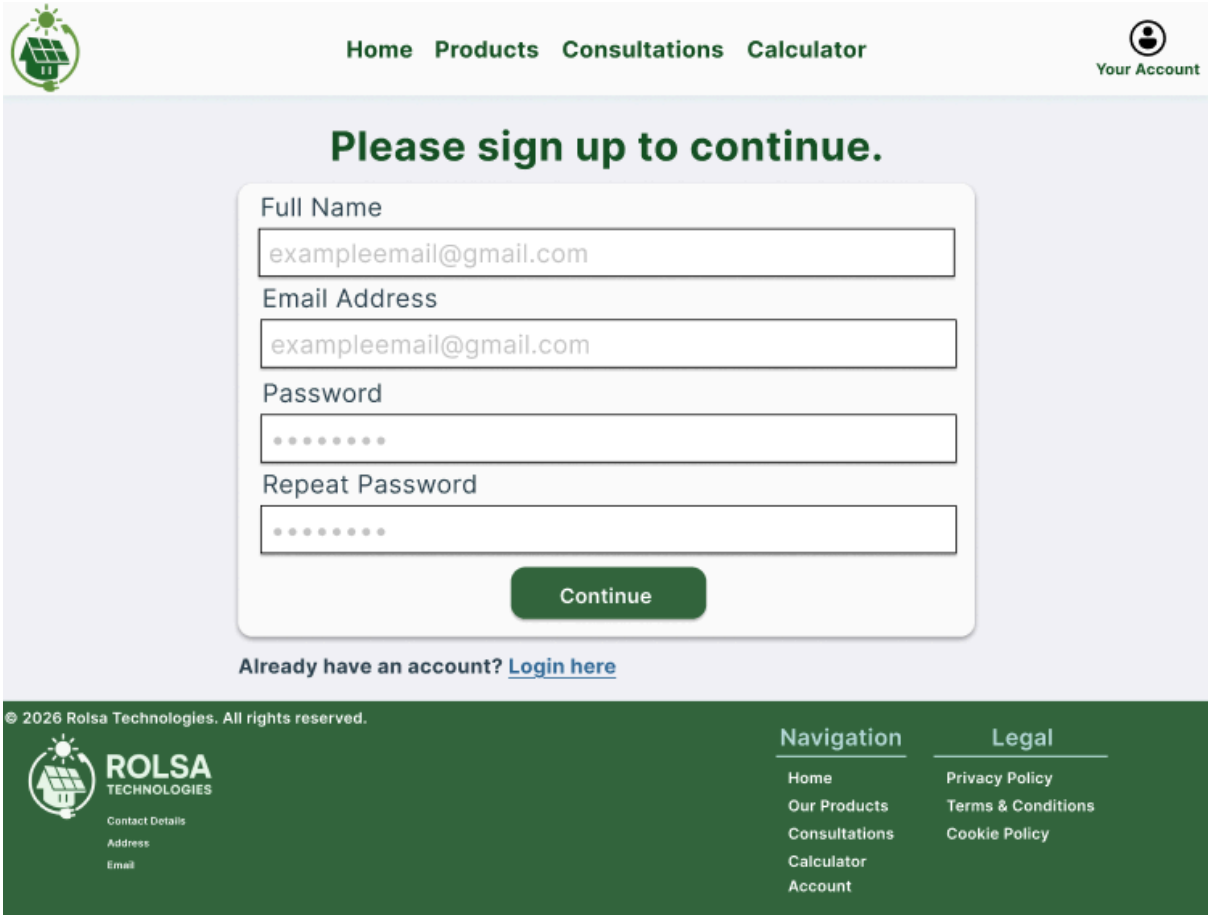
Logging in

This is a separate page from the main page. It is accessed by the “Your Account” button in the top right corner of the navigation bar.



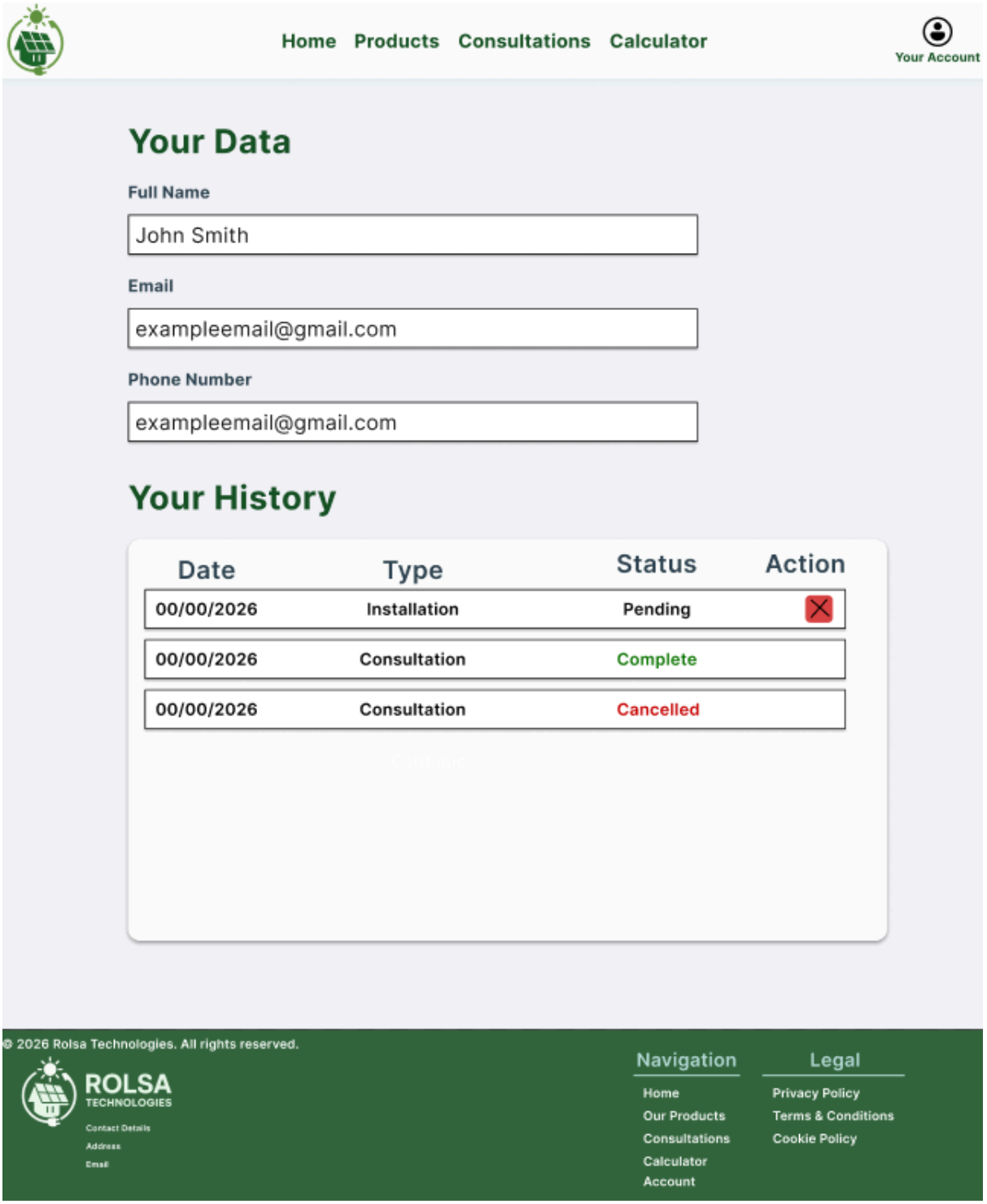
Signing Up

This is the signing up page, separate from both the “logging in” page and “your account” page. Accessed by pressing the “Sign up here” hyperlink within the log in page.

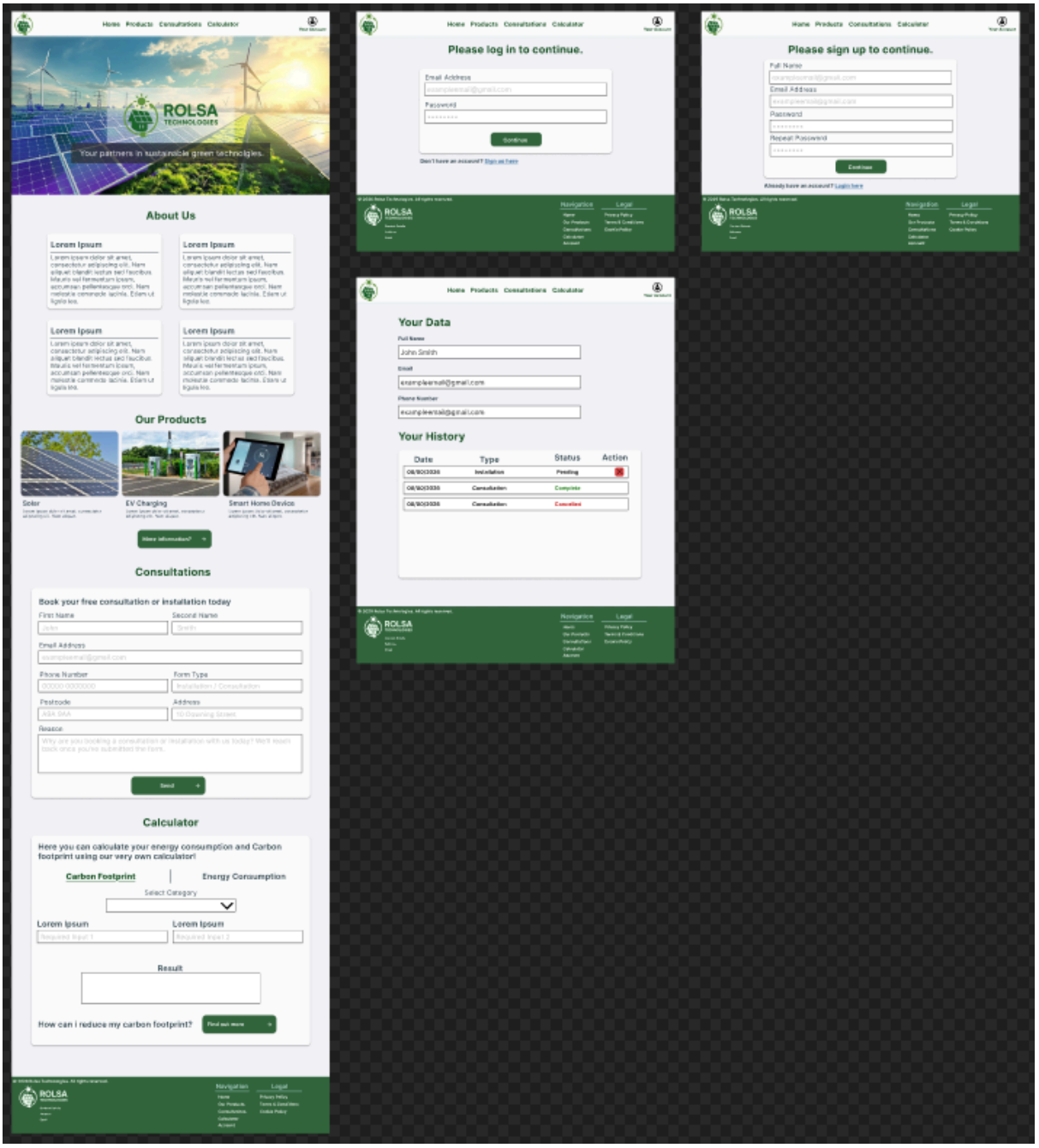


Manage Your Account

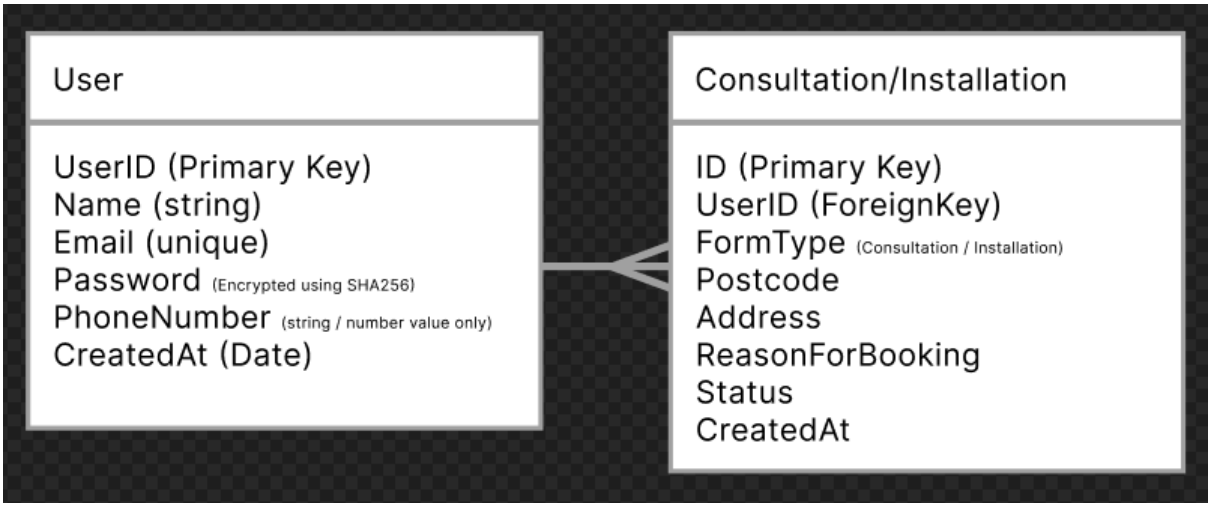
This page is separate from all the other pages. Accessed by either logging in, or pressing the “Your Account” button whilst being actively logged into an account.


Home
Products
Consultations
Calculator
Your Account
Your Data
Full Name
John Smith
Email
exampleemail@gmail.com
Phone Number
exampleemail@gmail.com
Your History
Date
Type
Status
Action
00/00/2026
Installation
Pending
00/00/2026
Consultation
Complete
00/00/2026
Consultation
Cancelled
© 2026 Rolsa Technologies. All rights reserved.
ROLSA
TECHNOLOGIES
Contact Details
Address
Email
Navigation
Home
Our Products
Consultations
Calculator
Account
Legal
Privacy Policy
Terms & Conditions
Cookie Policy

Full Solution



Entity Relationship Diagram (ERD)



One user can have multiple Consultations / Installations.

Data Dictionary

Consultations / Installations

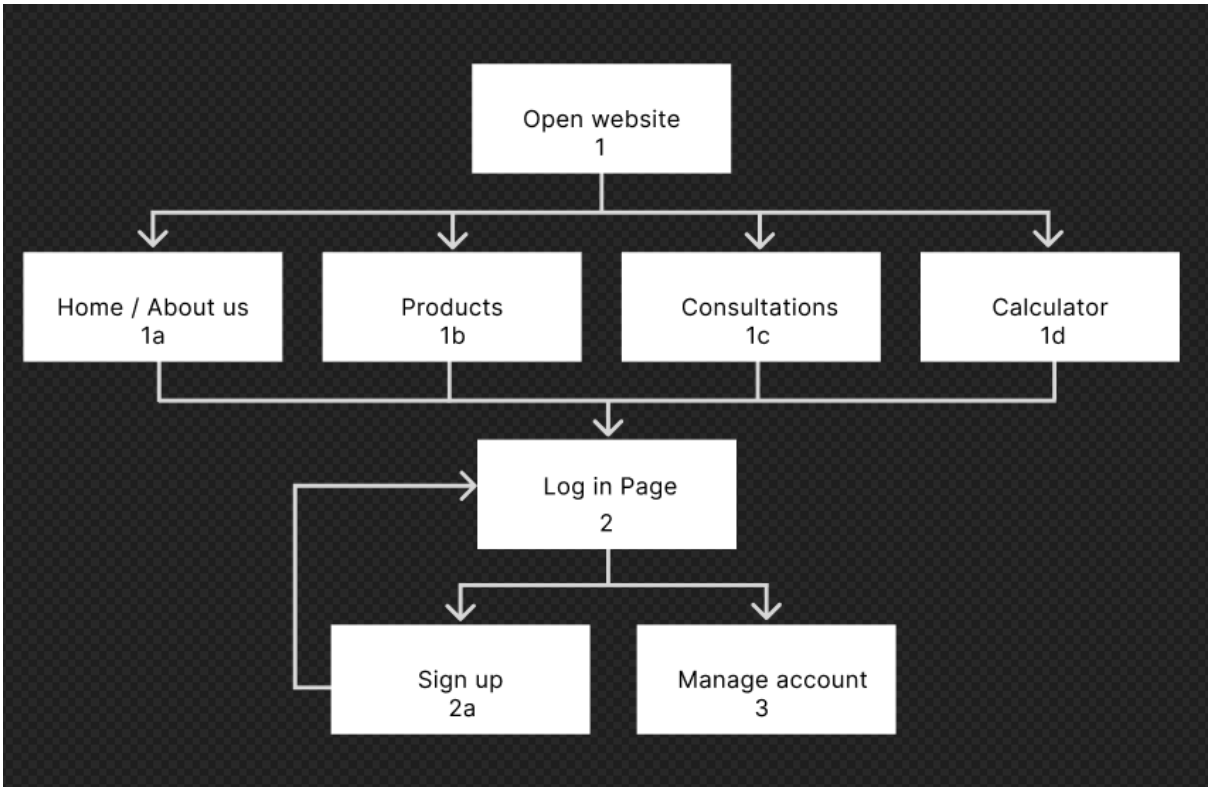
Field Name	Data Type	Constraints	Description
`ID`	INT	Primary Key, Auto Increment	Unique identifier for each booking
`UserID`	INT	Foreign Key -> User(UserID)	Links booking to the user who created it
`FormType`	VARCHAR(50)	NOT NULL	Type of form: Consultation or Installation
`Postcode`	VARCHAR(10)	NOT NULL	Postcode of the address
`Address`	VARCHAR(255)	NOT NULL	Full address for the booking
`ReasonForBooking`	TEXT	NULL	Reason provided by the user

`Status`	VARCHAR(50)	DEFAULT 'Pending'	Booking status (Pending, Confirmed, Completed)
`CreatedAt`	DATETIME	DEFAULT CURRENT_TIMESTAMP	When the booking was created

User

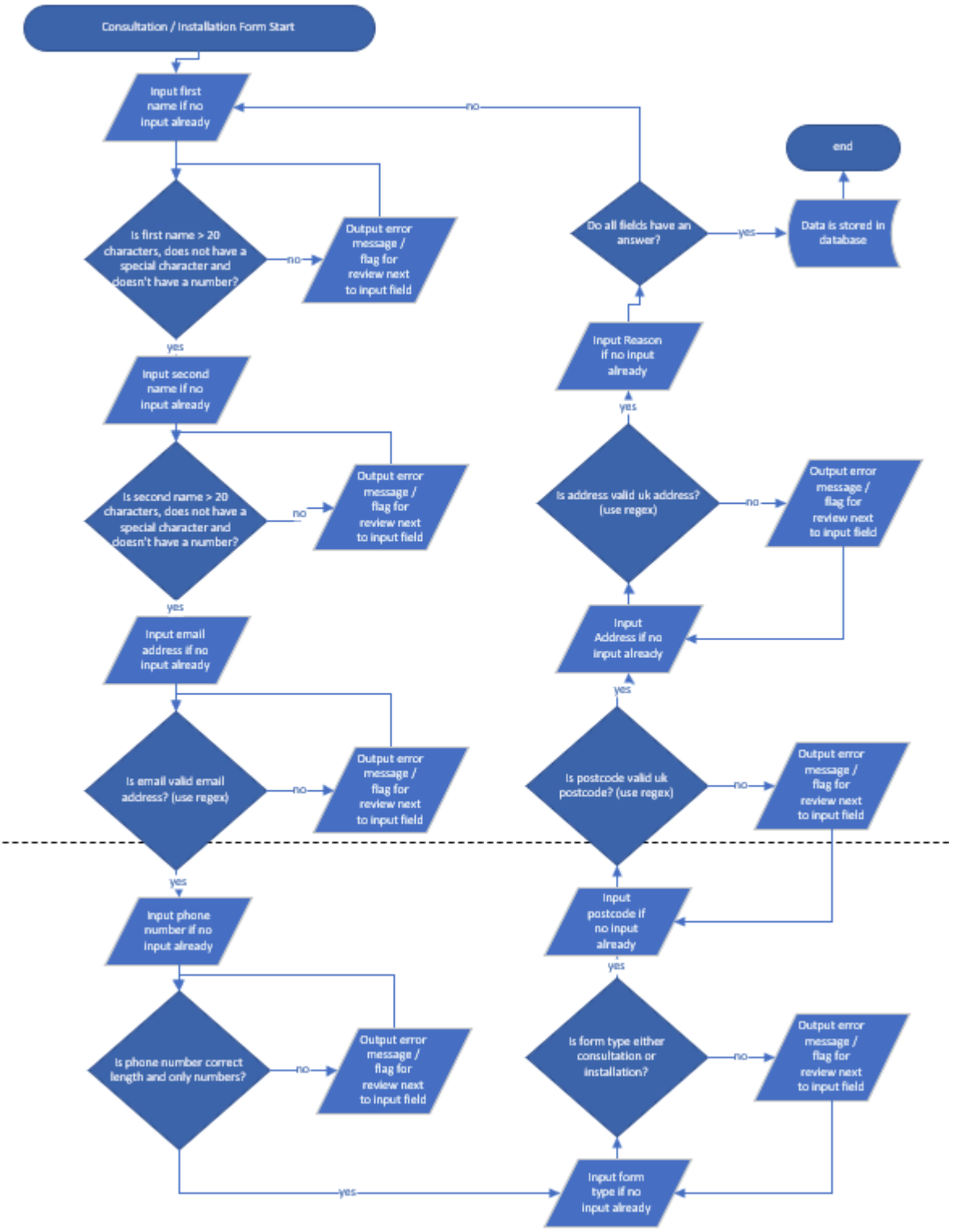
Field Name	Data Type	Constraints	Description
`UserID`	INT	Primary Key, Auto Increment	Unique identifier for each user
`Name`	VARCHAR(100)	NOT NULL	Full name of the user
`Email`	VARCHAR(255)	UNIQUE, NOT NULL	User's email address (used for login)
`Password`	CHAR(64)	NOT NULL	Encrypted password (SHA256 hash)
`PhoneNumber`	VARCHAR(15)	NULL	User's phone number
`CreatedAt`	DATETIME	DEFAULT CURRENT_TIMESTAMP	When the user account was created

Algorithm

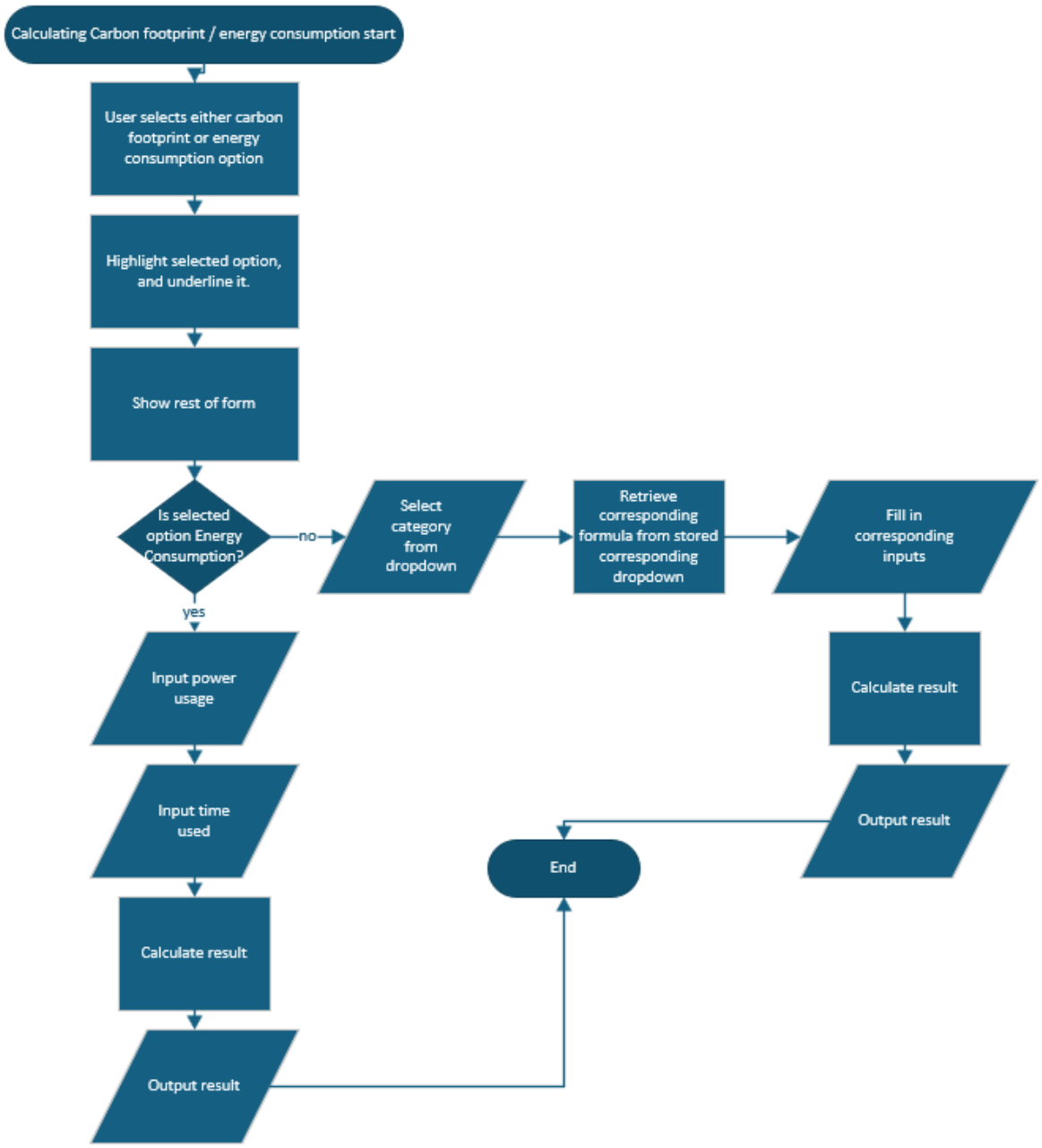

Open website
1
Home / About us
1a
Products
1b
Consultations
1c
Calculator
1d
Log in Page
2
Sign up
2a
Manage account
3

Flowcharts

The consultation / installation form has a simple form with just validation.



Calculations for energy consumption and carbon footprint vary, formulae will be stored and depending on the category selected, the corresponding formula will be applied, user will have to fill in the inputs



Test Strategy

Date of test	Component to be tested	Type of test to be carried out	Prerequisites and dependencies	Test
22/01/2026	Navigation Bar	Functionality	Need to have created the sections for each link, including a page for the accounts page for a redirect.	The developer will go through each of the hyperlinks present in the navigation bar (total of 5, 4 section links and accounts page link), ensuring that each link takes the user to the designated spot either on the page or to an external page.
22/01/2026	Accounts page: Logging in / Signing up	Functionality, Integration, Security	Database created, redirects to accounts page working, sign up / login pages are created with forms and correct php redirects	The developer will test the sign up process first. The developer will ensure that when the user presses the enter key or submit, the form checks and validates all the inputs making sure that the full name is no longer than 50 characters (25 for first, 25 for last), that the email address provided is in a valid format, and will check that the first password input is equal to the second password input, that the passwords are longer than 8 characters and checks if the password contains at least one of each special character and number. This is assuming that the user has not logged in prior. After this, the developer will check the backend data to ensure that the password has been securely hashed using sha256. This can be done through viewing the data in PHPMyAdmin. After this, the developer will ensure that the logging in system will work. This will be done by inputting the same email and passwords that were used when signing up. If the password and email matches the one provided in the database / signup process, the user should be redirected to the managed account page with a session being started. This is where integration testing is started as when the user goes back to the main page, and then

				tries to go back to the accounts page, it should see that a session is active and keep the user logged in. The developer should also try sad paths, where the wrong credentials are typed in. This should result in an error message being shown and the user is asked to try again if the credentials don't match that of which is in the system. SQL injections should also be tested or session hijacking, this shouldn't be possible as PDO (php) filters out sql injections.
22/01/2026	Consultation / Installation booking	Functionality, Integration, Security	Accounts pages fully working, full front end for consultations at a minimum	The developer will go to the consultations / installations form, where they will fill out the form with the corresponding details. Both maximum and minimum values are to be tried. Whilst entering values, the system should compare what the developer is entering with what is allowed. For example, if the user writes over 20 characters for the first name, an error message will show up saying that the input has exceeded the maximum character length. This validation will occur for all fields, with regex for emails, postcodes, addresses and phone numbers and the rest having standard procedure checks such as length checks or presence checks. Once the user presses submit, the form should check all fields and ensure that none are empty. If they are, the form will stop the submission and ask for the remaining fields to be filled in. Once this has been confirmed, the form is sent off. The developer should now access the accounts page, and should be logged in under the SAME email that was provided in the consultation / installation booking information. Under the history section, the user should be able to see their booking with status being pending, and the date they booked it and the type they chose. There should also be an action button that allows the consultation to be cancelled, which will remove its entry from the database.

				This can be checked manually through PHPMyAdmin.
22/02/2026	Calculator Testing	Functionality	Front end and backend of the calculator.	The developer will start by selecting from one of two options. Either calculating carbon footprint or energy consumption. Once the user presses either one, the selection option will turn green and become underlined. If the other one is pressed, it swaps and vice versa. Following this, if the user selected the energy consumption option, two simple input fields will appear + a results output. The user should be able to fill in the two fields with ONLY numbers, testing all characters and ensuring that only numbers can be inputted. The first value will be power consumption, the other will be duration. The only check for this one will be that the values provided are greater than 0 as you can not have negative time or negative power consumed. These values are then calculated to give a total energy consumption over the time provided. Alternatively, selecting the carbon footprint calculator, the user will be able to select from a dropdown menu the category of carbon footprint emissions they wish to calculate (transport, food etc.). This is important as the calculations are different for each category. Depending on what the user selects, the system will display different inputs with validation for only numbers that are greater than or equal to 0 being allowed. No other characters will / should be accepted (sad paths to be tested too). The calculator will perform the calculations based on the inputs provided allowing the user to see their carbon footprint. Lastly, there's a button at the bottom of the calculator that asks the user if they wish to

				have information regarding how they can decrease their carbon footprint. Once prompted, a UI element will appear on the users screen that will provide information about how to lower their carbon emissions. The only input allowed for this is the one to close it (the x button found on the top right of it), therefore, no other testing is required.
22/01/2026	Testing device compatibility	Functionality / Experience, Integration	Full solution front end needs to have been developed on one platform (such as PC)	The developer will emulate the website on a mobile device and ensure that the solution scales properly. All text should be easy to read and there should be no overlapping. Images should properly scale. If the boxes would be too small due to having to squeeze to fit 3 horizontally, they would instead stack on top of each other, creating a vertical stack instead of horizontal one. All features should still be easy to use with nothing being hidden or broken / not scaled as expected. Re-run the same tests to ensure that nothing breaks regardless of device.